Multi-Domain Few-Shot Medical Image Segmentation: An Empirical Study of UniverSeg Across Six Medical Imaging Domains

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Independent Study Project : Medical Image Segmentation

Code: `https://github.com/Manthan0000/UniverSeg_2.0.git`

Abstract

Abstract : *Deep learning models for medical image segmentation typically generalize poorly to unseen anatomies, imaging modalities, or label definitions, and adapting them to a new task usually requires retraining or fine-tuning a process that is often infeasible for clinical researchers without machine learning expertise. UniverSeg addresses this by reframing segmentation as a support-query problem: given a query image and a small support set of example image-label pairs that define a task, a single trained network predicts the query's segmentation mask without any additional training, using a CrossBlock mechanism to let query and support representations interact at multiple scales. In this work, we reproduce the UniverSeg pipeline end-to-end, train our own UniverSeg model from scratch on a reduced collection of six medical imaging datasets spanning brain MRI, cardiac MRI, cardiac ultrasound, retinal fundus imaging, dental X-ray, and abdominal CT , and compare it against the officially released pretrained checkpoint. We further study two properties of the support-query framework that are central to its practical use: the effect of support-set size on segmentation quality, and the sensitivity of predictions to which particular support examples are chosen. Our results confirm that larger support sets consistently improve Dice scores for both models, but reveal that this benefit scales very differently with training scale: the official checkpoint, trained on the full 53 dataset MegaMedical corpus, keeps improving up to a support size of 32 (reaching 0.9383 mean Dice), while our reduced-training model plateaus near 0.81 beyond a support size of roughly 8. We additionally show that segmentation quality is not determined solely by support-set size the specific composition of the support set can shift Dice scores measurably even when the size is held fixed. We report mean, best, and worst Dice scores, along with per-dataset sample and query counts, across all experiments and discuss the implications for support-set curation and training scale in real deployments of universal segmentation models.*

Index Terms : *medical image segmentation, few-shot learning, UniverSeg, cross-attention, support set, Dice score*

I. Introduction

Medical image segmentation is a central task in clinical image analysis, underlying applications ranging from tumor delineation to organ-volume estimation. Deep convolutional networks are the dominant tool for this task, but a network trained for one anatomy, modality, or label set generally fails to generalize to a new one because of domain shift. In practice this means that every new segmentation task a new organ, a new imaging device, a new patient population requires collecting labeled data and training or fine-tuning a dedicated model, which is expensive and requires expertise that most clinical researchers do not have.

UniverSeg [1] was proposed as a way to sidestep this problem entirely. Rather than learning a task specific mapping from image to mask, UniverSeg learns a single universal function that takes as input a query image together with a small support set of labeled example pairs describing the desired segmentation task, and predicts the query mask in a single forward pass, with no gradient updates at inference time. The key architectural novelty is the CrossBlock, a building block that lets the query representation and the representations of every support example interact through cross-convolutions, so that the network can transfer the notion of what to segment from the support examples onto the query image.

The goal of this project was twofold. First, we wanted to reproduce the UniverSeg training and evaluation pipeline ourselves building the dataset loaders, the preprocessing, the episodic task-sampling procedure, and the training loop and train a UniverSeg model from scratch on a smaller, six dataset collection rather than the full 53 dataset MegaMedical corpus

used by the original authors. Second, we wanted to go beyond simple reproduction and study two practical questions about the support-query mechanism that the original paper touches on but that are worth re examining on our own reduced setup: how much does the number of support examples matter, and how much does the specific choice of support examples matter when the size is fixed. These two extensions form the core empirical contribution of this report, in addition to a direct comparison between our from scratch model and the officially released pretrained checkpoint.

II. Related Work

Medical image segmentation has traditionally relied on architectures such as U-Net and its many variants, trained in a fully supervised, task-specific manner. Automated pipelines such as nnU-Net [2] remove much of the manual design effort but still require full retraining for every new task. Multitask learning frameworks can share a backbone across several tasks, but the set of tasks must be fixed at training time; a network trained this way cannot solve a task it has not seen. Transfer learning and meta-learning approaches reduce the amount of data or gradient steps needed for a new task, but they still require some fine tuning, which reintroduces the need for machine learning expertise and compute.

Few-shot segmentation methods, both in natural and medical imaging, are the closest relatives of UniverSeg : they predict a mask for a query image from a handful of labeled support examples without additional training. Prior medical few shot methods, however , were generally designed for a narrow notion of few-shot new classes within the same anatomy or modality rather than for generalizing across anatomies and modalities simultaneously. UniverSeg targets this broader setting directly by training on a highly diverse collection of segmentation tasks (the MegaMedical corpus, 53 datasets and over 22000 scans) so that the CrossBlock mechanism learns a general purpose notion of support to query transfer rather than one tied to a particular domain.

III. Base Model: UniverSeg

The base architecture used in this project is UniverSeg, an approach for universal medical image segmentation. Structurally, UniverSeg follows the general encoder–decoder idea of U-Net [3]:

$$\text{Image} \rightarrow \text{Encoder} \rightarrow \text{Feature Extraction} \rightarrow \text{Decoder} \rightarrow \text{Mask}$$

The key difference from a standard U-Net is that UniverSeg does not learn from a fixed set of images and masks alone. Instead, it follows a support query formulation. At inference (and at every training step), the model receives three inputs:

- A *query image* : the image to be segmented.
- A set of *support images* : example images from the same task.
- A set of *support masks* : the ground-truth segmentations for those support images

The support images and masks specify what the target structure looks like; the model then uses this information to predict the segmentation mask of the query image , rather than depending on the query image alone:

$$\text{Support Images + Masks} \xrightarrow{\text{CrossBlocks}} \text{Predicted Mask} \leftarrow \text{Query Image}$$

A. CrossBlock

CrossBlocks are the key component of UniverSeg. A CrossBlock lets the feature representation of the query image interact with the feature representation of every support example through a cross convolution operation, which concatenates the query features with each support example's features (channel wise) and passes the result through a shared convolution. Because the same convolutional weights are reused across all support examples , this operation is permutation-invariant with respect to the order of the support set, and it naturally supports a variable number of support examples. Stacking CrossBlocks at multiple encoder/decoder resolutions, in a U-Net like structure with skip connections, allows information to be exchanged between the support set and the query at multiple spatial scales. This is what allows UniverSeg to understand the relationship between what the target structure looks like in the support images and where a similar structure appears in the query image, and is what makes it suitable for segmentation across very different medical datasets and domains without retraining.

B. Input Format

The tensor shapes used in our implementation were:

- Query image: $(B, 1, 128, 128)$
- Support images: $(B, S, 1, 128, 128)$
- Support masks: $(B, S, 1, 128, 128)$

where B is the batch size and S is the support-set size. Our main training experiments used a batch size of 1, a support size of 4, and an image resolution of 128×128.

IV. Implementation

We implemented the full UniverSeg pipeline ourselves, using the official UniverSeg architecture , and evaluated it both with the officially released pretrained checkpoint and with our own weights trained from scratch The end-to-end pipeline used throughout this project is :

$$\text{Raw Dataset} \rightarrow \text{Dataset Loader} \rightarrow \text{Preprocessing} \rightarrow \text{Image + Mask} \rightarrow 128 \times 128 \text{ Tensor} \rightarrow \text{Support Set + Query Image} \rightarrow \text{UniverSeg / CrossBlocks} \rightarrow \text{Prediction Mask} \rightarrow \text{Dice Score} \rightarrow \text{Mean / Best / Worst Dice}$$

A. Datasets

For our reduced, multi domain implementation we used six medical imaging datasets, chosen to span a broad range of anatomies and imaging modalities (Table 1).

A seventh dataset , CHASE (retinal vessel imaging) , was initially prepared alongside the six datasets above but was excluded from the final, clean training run reported in this doc-

Table 1: Datasets used and their imaging domain / modality.

Dataset	Domain / Modality
OASIS	Brain MRI
ACDC	Cardiac MRI
DRIVE	Retinal fundus images
CAMUS	Cardiac ultrasound
PanDental	Dental panoramic X-ray
CHAOS	Abdominal CT / liver segmentation

ument, so all results below reflect training on OASIS, ACDC, DRIVE, CAMUS, PanDental, and CHAOS only.

B. Preprocessing

Different medical datasets arrive in different image sizes and formats, so every dataset was converted into a common representation before being used by the model. For each sample: the image was loaded; the corresponding segmentation mask was loaded; the image was resized to 128×128; the mask was resized using nearest-neighbor interpolation (to avoid introducing non-binary label values); the image was converted to a single channel; pixel intensities were normalized to the range $[0, 1]$; and masks were converted into binary masks.

C. Task Sampling

UniverSeg is trained with episodic support–query sampling. For every training task the following steps are performed:

1. Select one dataset.
2. Select support examples from that dataset.
3. Select one query image (held out from the chosen support examples).
4. Pass the support images and masks to the model.
5. Pass the query image to the model.
6. Compare the predicted query mask against the ground-truth query mask.
7. Compute the loss and update the model.

D. Training Settings

E. Loss Function

We used the Soft Dice Loss for training. The Dice coefficient measures the overlap between a predicted segmentation and the ground-truth segmentation:

$$\text{Dice} = \frac{2 \times |\text{Prediction} \cap \text{Ground Truth}|}{|\text{Prediction}| + |\text{Ground Truth}|} \tag{1}$$

$$\mathcal{L}_{\text{SoftDice}} = 1 - \text{Dice} \tag{2}$$

A lower loss corresponds to greater overlap between the predicted mask and the ground truth.

F. Final Trained Model

Our final model was trained on all six datasets with a support size of 4, batch size of 1, 5000 training steps, Soft Dice Loss, the Adam optimizer , and a learning rate of 1×10^{-4}. The exact configuration used to produce the final checkpoint is summarized in Table 2.

Table 2: Final training configuration and checkpoint location for our from-scratch UniverSeg model.

Hyperparameter	Value
Batch size	1
Support size	4
Training steps	5000
Learning rate	1×10^{-4}
Optimizer	Adam
Loss	Soft Dice Loss
Image resolution	128×128
Model	UniverSeg, trained from scratch
Checkpoint	`universeg_5000_steps.pth`

Table 3: Per-dataset sample counts, number of evaluation queries, and mean Dice score for the official pretrained checkpoint versus our 5000-step model.

Dataset	Samp.	Off.	Ours	Qry.
OASIS	414	0.8635	0.8109	20
ACDC	456	0.6610	0.8167	20
DRIVE	20	0.3608	0.2325	16
CAMUS	400	0.7569	0.7755	396
PanDental	116	0.8426	0.8514	112
CHAOS	2341	0.6640	0.5974	2337

V. Extension 1 Officially Pretrained Model vs Our Model

The first extension compares the officially released, pretrained UniverSeg checkpoint against our independently trained model, without retraining the official model in any way. Both models were evaluated on the same six datasets under identical conditions , and Dice scores were computed for each

Mean, best, and worst Dice were used as evaluation metrics. The purpose of this comparison is to understand how a reduced training setup six datasets and 5000 steps, versus the authors' full MegaMedical training regime affects cross-domain segmentation quality.

A. Results

The full evaluation, including the number of samples available in each dataset and the number of query images actually used for evaluation, is shown in Table 3 and Figure 1.

B. Discussion

The results show that relative performance varies considerably across medical domains, and neither model dominates the other uniformly. Our model, despite being trained on only six datasets for 5000 steps, outperformed the official checkpoint on ACDC, CAMUS, and PanDental , while the official checkpoint performed better on OASIS, DRIVE, and

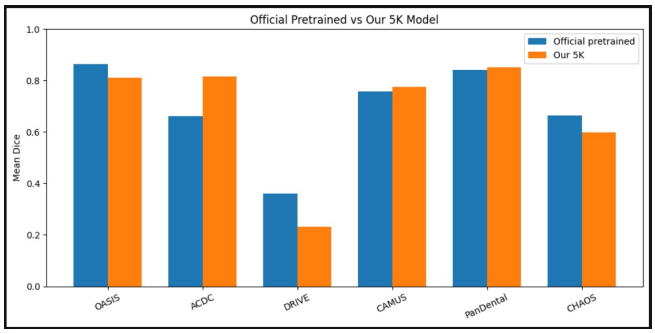


Figure 1: Mean Dice score per dataset for the official pretrained UniverSeg checkpoint versus our model trained for 5000 steps on six datasets.

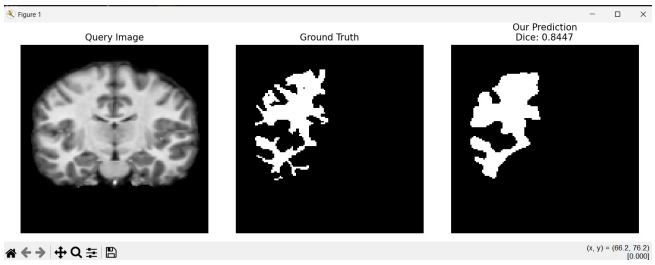


Figure 2: Example query image, ground-truth mask, and predicted mask for a coronal brain MRI slice (Dice = 0.8447).

CHAOS. Since our model was trained with only a reduced set of six datasets, whereas the official UniverSeg checkpoint was trained by the original authors on the much larger and more diverse 53-dataset MegaMedical corpus, this comparison illustrates the effect of training scale and dataset diversity on cross-domain generalization: broader training data appears to help most on tasks such as DRIVE (thin vessel structures) and CHAOS (abdominal organs), which are visually and structurally distant from the other datasets in our reduced collection, whereas for tasks closer to the datasets we trained on directly, our smaller model was competitive or better.

It is also worth noting the very different evaluation-set sizes across datasets (Table 3): CAMUS (396 queries) and CHAOS (2337 queries) give a statistically stable estimate of mean Dice, whereas DRIVE has only 16 query images out of 20 total samples, and OASIS and ACDC are evaluated on 20 queries each. The low scores on DRIVE for both models (0.3608 official, 0.2325 ours) should therefore be read with the caveat that they come from a very small and difficult dataset of thin, low-contrast retinal vessels, a structure type that is inherently hard for a 128×128, single-slice segmentation setup to capture accurately. Qualitative predictions were also visualized to compare the segmentation masks side by side (Figures 2–7).

VI. Extension 2 : Effect of Support-Set Size

A central property of UniverSeg is its use of a support set: a collection of example image–mask pairs that tells the model what kind of structure it should segment. In this extension

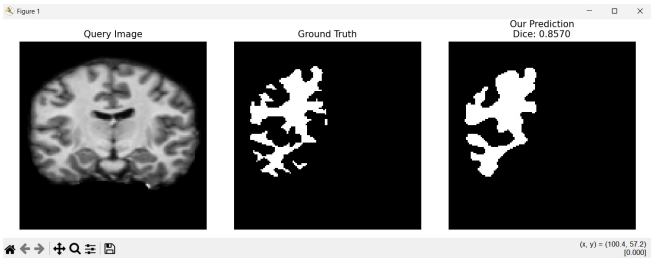


Figure 3: A second brain MRI segmentation example, showing a different slice and label region (Dice = 0.8570).

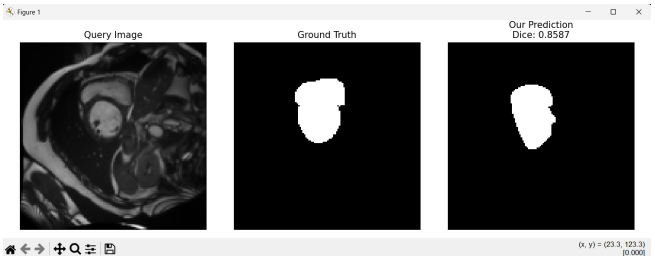


Figure 4: Cardiac MRI short-axis segmentation example (Dice = 0.8587).

we investigated how the number of support examples affects segmentation performance, while holding the query images and the model fixed. Support sizes tested: 1, 2, 4, 8, 16, 32, and 64.

A. Experimental Procedure

1. Select the required number of support image–mask pairs.
2. Provide them to UniverSeg alongside the query image.
3. Run segmentation on the same fixed set of query images for every support size.
4. Compute Dice scores and their mean across the query set.

We ran this sweep on both our from-scratch model and the officially released pretrained checkpoint, so that the effect of support-set size could also be compared against the effect of training scale seen in Extension 1.

B. Results

C. Discussion

Increasing the support-set size consistently improved segmentation performance for both models. For our model, the largest gains occur when moving from very small support sets (1–2 examples) to moderate ones (4 to 16 examples); beyond that point, additional support examples continue to help, but the marginal improvement becomes progressively smaller, plateauing around 0.81 Dice from a support size of 8 onward. This mirrors the intuition behind the CrossBlock design: a small support set gives the model a noisy, possibly unrepresentative picture of the target structure, whereas averaging information across more support examples lets the model build a more robust representation of the task before making its prediction on the query image.

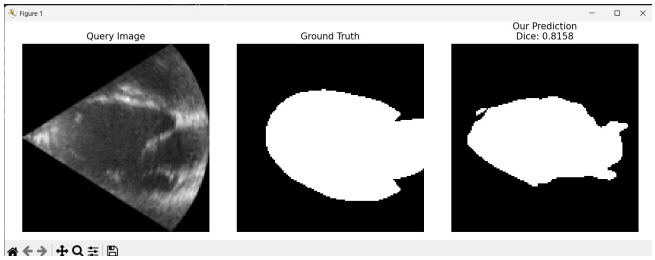


Figure 5: Cardiac ultrasound (echocardiography) segmentation example, showing the characteristic sector-shaped field of view (Dice = 0.8158).

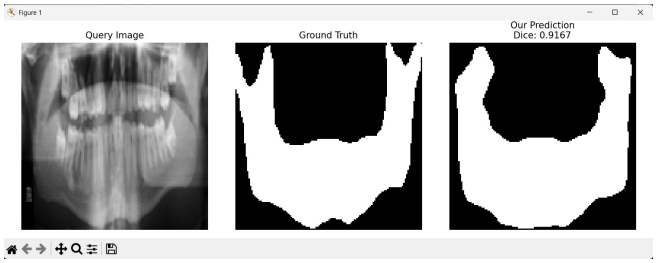


Figure 6: Dental panoramic X-ray mandible segmentation example (Dice = 0.9167).

The comparison with the official checkpoint reveals an important difference in degree rather than in trend. The official model starts lower at support size 1 (0.6627 versus our 0.7681) but rises much more steeply and continues improving well past the point where our model has plateaued, reaching 0.9045 at support size 16 and peaking at 0.9383 at support size 32 before a slight dip at 64. Our model, by contrast, gains comparatively little beyond a support size of 8 and never exceeds roughly 0.81. This suggests that the ability to keep extracting useful signal from a growing support set is itself something the network has to learn from a large and diverse training distribution: the official checkpoint, trained on the full 53-dataset MegaMedical corpus, appears to have learned a CrossBlock aggregation behavior that scales much more effectively with support-set size than the aggregation behavior learned by our model from only six datasets and 5000 steps. In other words, support-set size and training scale are not independent factors — a more thoroughly trained model is able to convert a larger support set into a substantially larger accuracy gain, whereas our reduced-training model saturates early regardless of how many support examples it is given.

VII. Extension 3 : Support-Set Selection Sensitivity

The third extension investigates whether the identity of the particular support examples selected for UniverSeg as opposed to merely their number affects the final segmentation performance. The comparison isolates this single factor:

> Same model + Same query images + Same support size
> − Different support examples

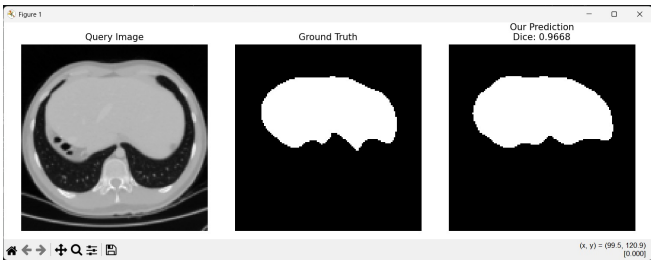


Figure 7: Abdominal/thoracic CT segmentation example (Dice = 0.9668).

Table 4: Mean Dice score as a function of support-set size, for the official pretrained checkpoint and for our 5000-step model.

Support Size	Official	Ours
1	0.6627	0.7681
2	0.8361	0.7889
4	0.8635	0.8037
8	0.8821	0.8061
16	0.9045	0.8083
32	0.9383	0.8113
64	0.9310	0.8143

A. Experimental Setup

- Support size fixed at 4.
- Query images: indices 100 to 119.
- Comparison A: randomly selected support examples.
- Comparison B: a curated support set obtained from a separate support-search procedure.

B. Results

Table 5: Mean Dice score for random versus curated (searched) support sets at a fixed support size of 4.

Support-Set Selection	Mean Dice
Random support	0.8109
Selected (curated) support	0.8020

C. Discussion

The curated support set did not outperform the randomly selected one if anything, it scored slightly lower on this query set. This is a useful negative result: it shows that UniverSeg's performance can be sensitive to the specific composition of the support set, and that a support-selection heuristic which appears reasonable a priori does not automatically translate into a higher Dice score. We do not conclude from this single comparison that any particular support-selection strategy is universally better or worse; the sample of query images (100–119) is small, and the gap between the two conditions (0.0089 Dice) is modest relative to the variability we observed in other experiments. Rather, this experiment demonstrates that support-set sensitivity is a real and measurable

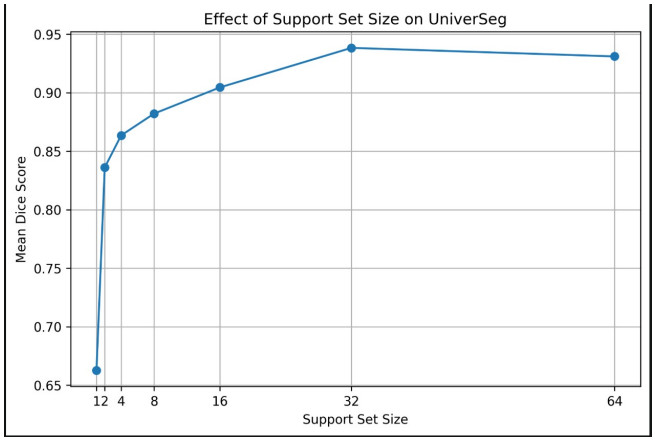


Figure 8: Mean Dice score as a function of support-set size, evaluated with the official pretrained UniverSeg checkpoint. Performance rises sharply up to a support size of 32 before a slight dip at 64.

phenomenon in the UniverSeg framework, and that the choice of support examples is not a neutral implementation detail it is a variable that can influence inference quality on par with, and sometimes more unpredictably than, the size of the support set studied in Extension 2. This reinforces the original UniverSeg paper's own observation that ensembling across several independently sampled support sets tends to reduce this variance, and suggests that principled support-set curation is a promising direction for future work.

VIII. Complete Project Pipeline

The complete project pipeline, from raw datasets to final experimental results, is summarized below:

> MEDICAL DATASETS → Dataset-specific Loaders → Preprocessing → 128 × 128 Image + 128 × 128 Mask → Task Sampling → [Support Set (Image + Mask × 4) + Query Image] → UniverSeg (Encoder → CrossBlocks → Decoder) → Predicted Mask → Dice Score → Mean / Best / Worst → Experiments (Official vs. Our Model; Support Size; Support Sensitivity) → Final Results

IX. Conclusion

In this project we reproduced the UniverSeg support query segmentation pipeline end-to-end and trained our own UniverSeg model from scratch on a reduced, six-dataset collection spanning brain MRI , cardiac MRI, cardiac ultrasound, retinal fundus imaging, dental X-ray, and abdominal CT. Comparing our model against the officially released pretrained checkpoint showed that relative performance depends heavily on how closely a held-out domain resembles the datasets seen during training, with neither model uniformly better across all six domains. Our two additional experiments characterized the practical behavior of the support-query mechanism: increasing support-set size steadily improves Dice scores with diminishing returns beyond roughly 8 to 16 examples, while the specific composition of a fixed-size support set can itself shift performance, showing that support-set curation is not a secondary detail but an active design choice in any deployment of a universal segmentation model. Together, these findings support UniverSeg's central premise that a single trained network can adapt to new medical segmentation tasks purely through the examples it is show while also highlighting that the practical benefit of this approach depends on both the amount and the quality of support information provided at inference time.

References

[1] V. I. Butoi, J. J. G. Ortiz, T. Ma, M. R. Sabuncu, J. Guttag, and A. V. Dalca, "UniverSeg: Universal medical image segmentation," *arXiv preprint arXiv:2304.06131*, 2023.

[2] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, no. 2, pp. 203–211, 2021.

[3] O. Ronneberger, P. Fischer, and P. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.

[4] D. S. Marcus, T. H. Wang, J. Parker, J. G. Csernansky, J. C. Morris, and R. L. Buckner, "Open Access Series of Imaging Studies (OASIS): Cross-sectional MRI data in young, middle aged, nondemented, and demented older adults," *Journal of Cognitive Neuroscience*, vol. 19, no. 9, pp. 1498–1507, 2007.

[5] O. Bernard, A. Lalande, C. Zotti, F. Cervenansky, X. Yang, P.-A. Heng, I. Cetin, K. Lekadir, O. Camara, M. A. G. Ballester, et al., "Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved?" *IEEE Transactions on Medical Imaging*, vol. 37, no. 11, pp. 2514–2525, 2018.

[6] J. Staal, M. D. Abràmoff, M. Niemeijer, M. A. Viergever, and B. van Ginneken, "Ridge-based vessel segmentation in color images of the retina," *IEEE Transactions on Medical Imaging*, vol. 23, no. 4, pp. 501–509, 2004.

[7] S. Leclerc, E. Smistad, J. Pedrosa, A. Ostvik, F. Cervenansky, F. Espinosa, T. Espeland, E. A. R. Berg, P.-M. Jodoin, T. Grenier, et al., "Deep learning for segmentation using an open large-scale dataset in 2D echocardiography," *IEEE Transactions on Medical Imaging*, vol. 38, no. 9, pp. 2198–2210, 2019.

[8] A. H. Abdi, S. Kasaei, and M. Mehdizadeh, "Automatic segmentation of mandible in panoramic X-ray," *Journal of Medical Imaging*, vol. 2, no. 4, p. 044003, 2015.

[9] A. E. Kavur, N. S. Gezer, M. Baris, S. Aslan, P.-H. Conze, V. Groza, D. D. Pham, S. Chatterjee, P. Ernst,

S. Ozkan, et al., "CHAOS challenge – combined (CT-MR) healthy abdominal organ segmentation," *Medical Image Analysis*, vol. 69, p. 101950, 2021.

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